

# Statistical Computing

Constrained Optimization & Duality

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Spring 2026

Constrained Optimization

Equality Constrained Optimization

Inequality Constrained Optimization

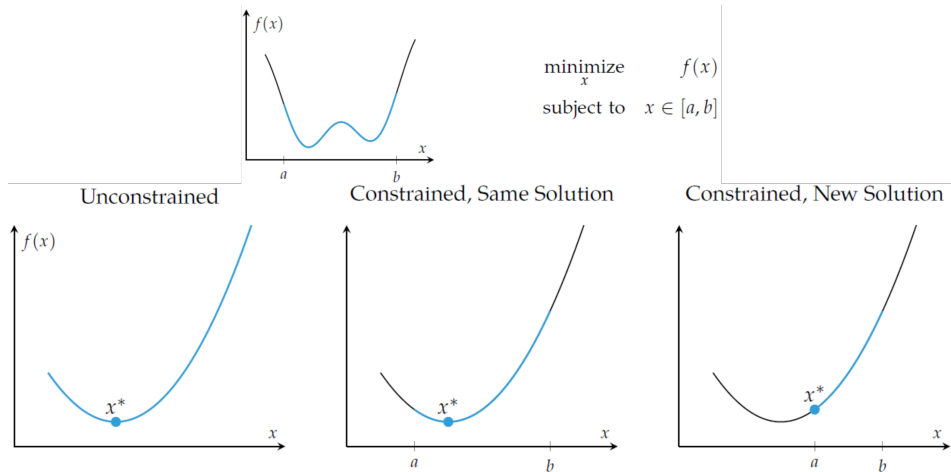
Lagrangian Duality

# Constrained Optimization

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# Constrained Maxima and Minima

- In many practical optimization problems, we must maximize or minimize a function in which the independent variables are subjected to certain further constraints.



# Equality Constrained Optimization

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# Substitution Method

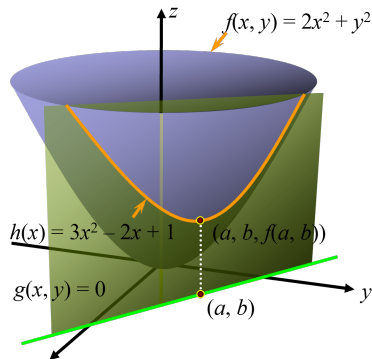
- Find the relative minimum of  $f(x, y) = 2x^2 + y^2$  subject to  $g(x, y) = x + y - 1 = 0$ .

## Solution.

Solving the constraint for  $y$  gives  $y = -x + 1$ . Substituting into  $f$ :

$$h(x) = 2x^2 + (1 - x)^2 = 3x^2 - 2x + 1.$$

Solve  $h'(x) = 6x - 2 = 0 \Rightarrow x = \frac{1}{3}$ . Since  $h''(x) = 6 > 0$ , this is a relative minimum. With  $y = \frac{2}{3}$ , the constrained minimum is at  $(\frac{1}{3}, \frac{2}{3})$ .



# The Method of Lagrange Multipliers

- To find the relative extrema of  $f(x, y)$  subject to  $g(x, y) = 0$ :

1. Form the **Lagrangian function** (where  $\lambda$  is the **Lagrange multiplier**):

$$F(x, y, \lambda) = f(x, y) + \lambda g(x, y).$$

2. Solve the system of equations

$$F_x = 0, \quad F_y = 0, \quad F_\lambda = 0$$

for all values of  $x$ ,  $y$ , and  $\lambda$ .

3. The solutions found in step 2 are candidates for the extrema of  $f$ .

## Example: Lagrange Multipliers

- Find the relative minimum of  $f(x, y) = 2x^2 + y^2$  subject to  $x + y = 1$  using Lagrange multipliers.

### Solution.

- Write  $g(x, y) = x + y - 1 = 0$ . Form the Lagrangian:

$$F(x, y, \lambda) = (2x^2 + y^2) + \lambda(x + y - 1).$$

- Solve the system:

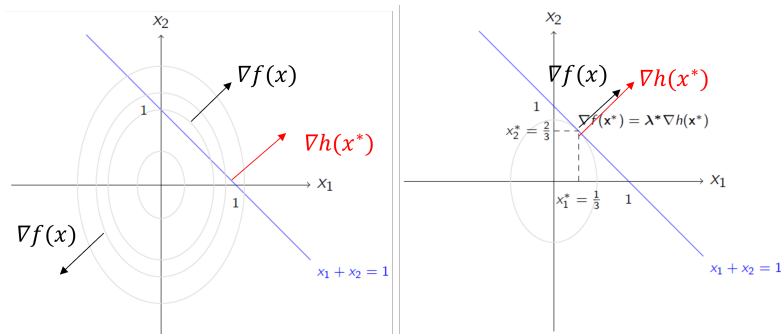
$$F_x = 4x + \lambda = 0, \quad F_y = 2y + \lambda = 0, \quad F_\lambda = x + y - 1 = 0.$$

- From the first two equations:  $x = -\frac{\lambda}{4}$  and  $y = -\frac{\lambda}{2}$ .
- Substituting into the third:  $\lambda = -\frac{4}{3}$ , so  $x = \frac{1}{3}$ ,  $y = \frac{2}{3}$ .



# Geometric Interpretation

$$\begin{array}{ll}\text{Minimize} & 2x_1^2 + x_2^2 \\ \text{Subject to} & x_1 + x_2 = 1\end{array}$$



## Geometric Interpretation

- Consider the gradients of  $f$  and  $h$  at the optimal point  $\mathbf{x}^*$ .
- They must be parallel (same direction, possibly different lengths):

$$\nabla f(\mathbf{x}^*) = \nu^* \nabla h(\mathbf{x}^*).$$

- This is equivalent to  $\nabla L(\mathbf{x}^*, \nu^*) = 0$  along with feasibility of  $\mathbf{x}^*$ .
- $\nabla f(\mathbf{x})$  is perpendicular to a level curve of  $f$ ; similarly  $\nabla h(\mathbf{x})$  is perpendicular to the level curve  $h(\mathbf{x}) = 0$ .
- From the example, at  $x_1^* = \frac{1}{3}$ ,  $x_2^* = \frac{2}{3}$ ,  $\lambda_1^* = \frac{4}{3}$ :

$$\nabla f(x_1^*, x_2^*) = \begin{bmatrix} \frac{4}{3} & \frac{4}{3} \end{bmatrix}, \quad \nabla h_1(x_1^*, x_2^*) = \begin{bmatrix} 1 & 1 \end{bmatrix}.$$

# Equality Constrained Optimization

- Optimization problem with one equality constraint:

$$\begin{aligned} & \text{minimize} && f(\mathbf{x}) \\ & \text{subject to} && h(\mathbf{x}) = 0. \end{aligned}$$

- The **Lagrangian** is  $L(\mathbf{x}, \nu) = f(\mathbf{x}) + \nu h(\mathbf{x})$ .
- A solution must satisfy  $\nabla_{\mathbf{x}} L(\mathbf{x}^*, \nu^*) = \nabla f(\mathbf{x}^*) + \nu^* \nabla h(\mathbf{x}^*) = \mathbf{0}$ , i.e., there exists  $\nu^*$  such that

$$\frac{\partial L(\mathbf{x}^*, \nu^*)}{\partial x_1} = \dots = \frac{\partial L(\mathbf{x}^*, \nu^*)}{\partial x_n} = 0, \quad \frac{\partial L(\mathbf{x}^*, \nu^*)}{\partial \nu} = h(\mathbf{x}^*) = 0.$$

## Equality Constrained Optimization: Two Constraints

- Optimization problem with two equality constraints:

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) \\ \text{subject to} & h_1(\mathbf{x}) = 0, \ h_2(\mathbf{x}) = 0.\end{array}$$

- The Lagrangian is  $L(\mathbf{x}, \boldsymbol{\nu}) = f(\mathbf{x}) + \nu_1 h_1(\mathbf{x}) + \nu_2 h_2(\mathbf{x})$ .
- Stationarity condition:

$$\nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\nu}) = \nabla f(\mathbf{x}) + \nu_1 \nabla h_1(\mathbf{x}) + \nu_2 \nabla h_2(\mathbf{x}) = \mathbf{0}.$$

## Equality Constrained Optimization: General Case

- **First-Order Necessary Conditions (equality constraints only):**
- General problem with  $m$  equality constraints:

$$\begin{aligned} &\text{minimize} && f(\mathbf{x}) \\ &\text{subject to} && h_j(\mathbf{x}) = 0, \quad j = 1, \dots, m. \end{aligned}$$

- Suppose  $\mathbf{x}^*$  is a local solution and LICQ holds at  $\mathbf{x}^*$ .
  - **LICQ** (Linear Independence Constraint Qualification): the constraint gradients  $\nabla h_1(\mathbf{x}^*), \dots, \nabla h_m(\mathbf{x}^*)$  are linearly independent.
- The Lagrangian is  $L(\mathbf{x}, \mathbf{v}) = f(\mathbf{x}) + \sum_{j=1}^m v_j h_j(\mathbf{x}) = f(\mathbf{x}) + \mathbf{v}^\top \mathbf{h}(\mathbf{x})$ .
- Then there exists  $\mathbf{v}^*$  such that at  $(\mathbf{x}^*, \mathbf{v}^*)$ :

$$\begin{aligned} \nabla_{\mathbf{x}} L(\mathbf{x}^*, \mathbf{v}^*) &= \nabla f(\mathbf{x}^*) + \sum_{j=1}^m v_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0}, \\ h_j(\mathbf{x}^*) &= 0, \quad j = 1, \dots, m. \end{aligned}$$

## LICQ: Why Linear Independence?

- **Example:**

$$\begin{array}{ll}\text{Minimize} & x_1 + x_2 + x_3^2 \\ \text{Subject to:} & x_1 = 1, \quad x_1^2 + x_2^2 = 1.\end{array}$$

The minimum is achieved at  $x_1 = 1, x_2 = 0, x_3 = 0$ .

The Lagrangian is:

$$L(x_1, x_2, x_3, \lambda_1, \lambda_2) = x_1 + x_2 + x_3^2 + \lambda_1(1 - x_1) + \lambda_2(1 - x_1^2 - x_2^2).$$

But  $\frac{\partial L(1, 0, 0, \lambda_1, \lambda_2)}{\partial x_2} = 1$  for all  $\lambda_1, \lambda_2$ : no multipliers can satisfy stationarity!

Observe:  $\nabla h_1(1, 0, 0) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$  and  $\nabla h_2(1, 0, 0) = \begin{bmatrix} 2 & 0 & 0 \end{bmatrix}$  are *linearly dependent*.

# Inequality Constrained Optimization

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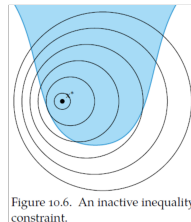
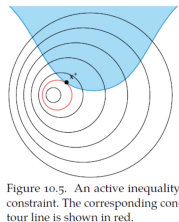
# Inequality Constrained Optimization

- Optimization problem with one inequality constraint:

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) \\ \text{subject to} & g(\mathbf{x}) \leq 0.\end{array}$$

- If the solution lies at the constraint boundary ( $g(\mathbf{x}^*) = 0$ ), the Lagrange condition holds:

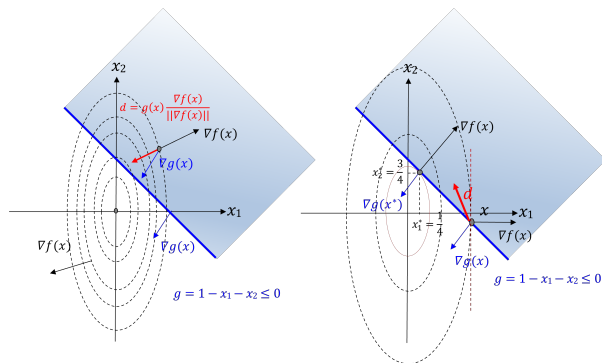
$$\nabla_{\mathbf{x}} L(\mathbf{x}^*, \lambda^*) = \nabla f(\mathbf{x}^*) + \lambda^* \nabla g(\mathbf{x}^*) = \mathbf{0}.$$





## Example: Inequality Constraint

$$\begin{aligned} \text{Minimize} \quad & 3x_1^2 + x_2^2 \\ \text{Subject to:} \quad & x_1 + x_2 \geq 1 \end{aligned}$$



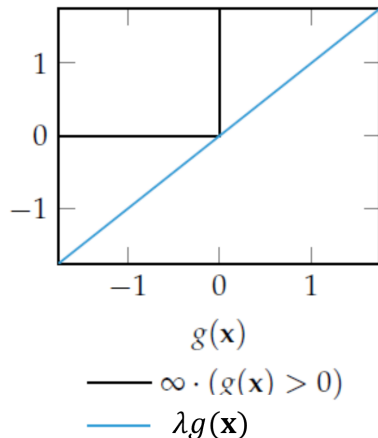
# Motivation for the KKT Conditions

- Imposing  $g(\mathbf{x}) \leq 0$  is equivalent to minimizing a step-penalized objective:

$$\min_{\mathbf{x}} f(\mathbf{x}) \text{ s.t. } g(\mathbf{x}) \leq 0 \iff \min_{\mathbf{x}} f_{\infty}(\mathbf{x}),$$

where  $f_{\infty}(\mathbf{x}) := f(\mathbf{x}) + \infty \cdot \mathbf{1}_{g(\mathbf{x}) > 0}(\mathbf{x})$  equals  $f(\mathbf{x})$  when  $g(\mathbf{x}) \leq 0$  and  $+\infty$  otherwise.

- We use a linear penalty  $\lambda g(\mathbf{x})$  ( $\lambda \geq 0$ ) as a lower bound on  $\infty \cdot \mathbf{1}_{g(\mathbf{x}) > 0}(\mathbf{x})$ , penalizing infeasibility.



## Motivation for the KKT Conditions (continued)

- **Lagrangian for inequality constraints:**

$$L(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda g(\mathbf{x}).$$

- Recover  $f_\infty$  by maximizing over  $\lambda$ :

$$f_\infty(\mathbf{x}) = \max_{\lambda \geq 0} L(\mathbf{x}, \lambda).$$

- **(Primal problem)**

$$\min_{\mathbf{x}} \max_{\lambda \geq 0} L(\mathbf{x}, \lambda).$$

## KKT Conditions: Feasibility and Slackness

- From the primal formulation  $\min_{\mathbf{x}} \max_{\lambda \geq 0} L(\mathbf{x}, \lambda)$ , KKT conditions emerge:
  - **Feasibility:**  $g(\mathbf{x}^*) \leq 0$ .
  - **Dual feasibility:**  $\lambda^* \geq 0$ .
  - **Complementary slackness:**  $\lambda^* g(\mathbf{x}^*) = 0$ .
    - A feasible point on the boundary ( $g(\mathbf{x}) = 0$ ) may have  $\lambda > 0$ ; a strictly feasible point ( $g(\mathbf{x}) < 0$ ) must have  $\lambda = 0$  to recover  $f(\mathbf{x}^*)$  from the Lagrangian.

- **Stationarity:**  $\nabla_{\mathbf{x}} f(\mathbf{x}^*) + \lambda^* \nabla_{\mathbf{x}} g(\mathbf{x}^*) = \mathbf{0}$ .
  - When the constraint is active, contour lines of  $f$  and  $g$  must be aligned. When inactive,  $\nabla f(\mathbf{x}^*) = \mathbf{0}$  and  $\lambda^* = 0$ .

$$\min_{\mathbf{x}} f(\mathbf{x}) \text{ s.t. } g(\mathbf{x}) \leq 0 \iff \min_{\mathbf{x}} \max_{\lambda \geq 0} L(\mathbf{x}, \lambda) := f(\mathbf{x}) + \lambda g(\mathbf{x})$$

# Theory of Constrained Optimization

- **Constrained optimization problem:**

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) \\ \text{subject to} & \mathbf{g}(\mathbf{x}) \leq \mathbf{0}, \quad \mathbf{h}(\mathbf{x}) = \mathbf{0},\end{array}$$

where  $f, g_i, h_j : \mathbb{R}^n \rightarrow \mathbb{R}$ .

- Feasible set:  $\Omega = \{\mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) \leq 0, i = 1, \dots, \ell, h_j(\mathbf{x}) = 0, j = 1, \dots, m\}$ .
- At a feasible point  $\mathbf{x}$ , inequality constraint  $i$  is **active** if  $g_i(\mathbf{x}) = 0$ , and **inactive** if  $g_i(\mathbf{x}) < 0$ .
- $\mathbf{x}^*$  is a local solution if  $\mathbf{x}^* \in \Omega$  and  $f(\mathbf{x}) \geq f(\mathbf{x}^*)$  for all  $\mathbf{x}$  in some neighborhood  $\mathcal{N}(\mathbf{x}^*) \cap \Omega$ .
- **Lagrangian:**

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^\top \mathbf{g}(\mathbf{x}) + \boldsymbol{\nu}^\top \mathbf{h}(\mathbf{x}).$$

- Active set:  $\mathcal{A}(\mathbf{x}) = \{i : g_i(\mathbf{x}) = 0\} \cup \{j : h_j(\mathbf{x}) = 0\}$ .

- **Why do we need LICQ?** Without it, degenerate behavior may prevent KKT conditions from holding:
  - E.g.,  $c_1(\mathbf{x}) = x_1^2 + x_2^2 - 2 = 0$  and  $c_2(\mathbf{x}) = (x_1^2 + x_2^2 - 2)^2 = 0$  give  $\nabla c_2(\mathbf{x}) = \mathbf{0}$  at every feasible point.
- **LICQ:** Given  $\mathbf{x}^*$  and active set  $\mathcal{A}(\mathbf{x}^*)$ , the set of active constraint gradients  $\{\nabla g_i(\mathbf{x}^*) : i \in \mathcal{A}(\mathbf{x}^*)\}$  is linearly independent.
  - If LICQ holds, none of the active constraint gradients can be zero.

# Karush–Kuhn–Tucker (KKT) Conditions

- **First-Order Necessary Conditions (KKT):** Suppose  $\mathbf{x}^*$  is a local solution and LICQ holds. Then there exist multipliers  $(\lambda^*, \nu^*)$  satisfying:

$$\nabla_{\mathbf{x}} L(\mathbf{x}^*, \lambda^*, \nu^*) = \nabla f(\mathbf{x}^*) + \sum_i \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_j \nu_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0} \quad (\text{Stationarity})$$

$$g_i(\mathbf{x}^*) \leq 0, \quad \forall i \quad (\text{Feasibility}), \quad h_j(\mathbf{x}^*) = 0, \quad \forall j \quad (\text{Feasibility}),$$

$$\lambda_i^* \geq 0, \quad \forall i \quad (\text{Dual feasibility}), \quad \lambda_i^* g_i(\mathbf{x}^*) = 0, \quad \forall i \quad (\text{Complementary slackness}).$$

- Inactive constraints ( $g_i(\mathbf{x}^*) < 0$ ) automatically have  $\lambda_i^* = 0$  by complementary slackness.



## Example: KKT with Inequality Constraints

$$\begin{array}{ll} \text{maximize} & -(x-2)^2 - 2(y-1)^2 \\ \text{subject to} & x+4y \leq 3, \quad x \geq y \end{array} \Rightarrow \begin{array}{ll} \text{minimize} & (x-2)^2 + 2(y-1)^2 \\ \text{subject to} & x+4y-3 \leq 0, \quad -x+y \leq 0. \end{array}$$

Lagrangian:  $L = (x-2)^2 + 2(y-1)^2 + \mu_1(x+4y-3) + \mu_2(-x+y)$ .

KKT stationarity:

$$\frac{\partial L}{\partial x} = 2(x-2) + \mu_1 - \mu_2 = 0, \quad \frac{\partial L}{\partial y} = 4(y-1) + 4\mu_1 + \mu_2 = 0.$$

Complementary slackness:  $\mu_1(x+4y-3) = 0, \quad \mu_2(-x+y) = 0, \quad \mu_1, \mu_2 \geq 0.$

## Example: KKT with Inequality Constraints (continued)

Two complementarity conditions  $\Rightarrow$  check 4 cases:

- **Case 1**  $\mu_1 = \mu_2 = 0$ : stationarity  $\Rightarrow x = 2, y = 1$ . But  $2 + 4(1) = 6 > 3$ : *infeasible*.
- **Case 2**  $\mu_1 = 0, x = y$ : stationarity  $\Rightarrow x = y = \frac{4}{3}$ , but  $\mu_2 = -\frac{4}{3} < 0$ : *dual infeasible*.
- **Case 3**  $x + 4y = 3, \mu_2 = 0$ : stationarity  $\Rightarrow x = \frac{5}{3}, y = \frac{1}{3}, \mu_1 = \frac{2}{3} > 0$ .  
Inactive constraint:  $-x + y = -\frac{4}{3} \leq 0 \checkmark$ . **Valid KKT point.**
- **Case 4**  $x + 4y = 3, x = y$ : stationarity  $\Rightarrow x = y = \frac{3}{5}$ , but  $\mu_2 = -\frac{48}{25} < 0$ : *dual infeasible*.

Optimal solution:  $x^* = \frac{5}{3}, y^* = \frac{1}{3}, (x^* - 2)^2 + 2(y^* - 1)^2 = 1$ .

## Example: KKT Conditions

$$\begin{array}{ll}\text{maximize} & -(x-3)^2 - (y-2)^2 \\ \text{subject to} & x^2 + y^2 \leq 5, \quad x + 2y \leq 4, \quad x, y \geq 0\end{array}$$

$$\nabla f = (6 - 2x, 4 - 2y), \quad \nabla g_1 = (2x, 2y), \quad \nabla g_2 = (1, 2), \quad \nabla g_3 = (-1, 0), \quad \nabla g_4 = (0, -1).$$

At  $(x, y) = (2, 1)$ : feasible with active set  $\mathcal{I} = \{1, 2\}$ . KKT stationarity:

$$\begin{bmatrix} 2 \\ 2 \end{bmatrix} - \mu_1 \begin{bmatrix} 4 \\ 2 \end{bmatrix} - \mu_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \mathbf{0} \quad \Rightarrow \quad \mu_1 = \frac{1}{3}, \quad \mu_2 = \frac{2}{3}.$$

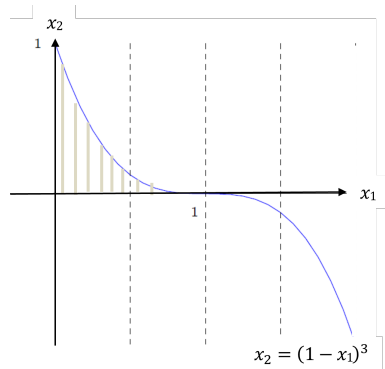
## Example: KKT with Multiple Constraints

$$\begin{aligned} \text{minimize} \quad & (x_1 - \tfrac{3}{2})^2 + (x_2 - \tfrac{1}{2})^2 \\ \text{subject to} \quad & c_1(\mathbf{x}) = x_1 + x_2 - 1 \leq 0, \\ & c_2(\mathbf{x}) = x_1 - x_2 - 1 \leq 0, \\ & c_3(\mathbf{x}) = -x_1 + x_2 - 1 \leq 0, \\ & c_4(\mathbf{x}) = -x_1 - x_2 - 1 \leq 0. \end{aligned}$$

- Solution:  $\mathbf{x}^* = (1, 0)^T$ ; active constraints:  $c_1, c_2$ .

$$\nabla f(\mathbf{x}^*) = \begin{bmatrix} -1 \\ -\frac{1}{2} \end{bmatrix}, \quad \nabla c_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \nabla c_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

- KKT multipliers:  $\lambda^* = (\frac{3}{4}, \frac{1}{4}, 0, 0)^T$ .



## ■ Class Exercise 1

$$\begin{aligned} & \text{minimize} && 2x_1^2 + 2x_1x_2 + x_2^2 - 10x_1 - 10x_2 \\ & \text{subject to} && x_1^2 + x_2^2 \leq 5, \quad 3x_1 + x_2 \leq 6. \end{aligned}$$

Solution:  $x_1^* = 1$ ,  $x_2^* = 2$ ,  $\lambda^* = 1$ .

## ■ Class Exercise 2

$$\begin{aligned} & \text{maximize} && \ln x + 1 + y \\ & \text{subject to} && 2x + y \leq 3, \quad x, y \geq 0. \end{aligned}$$

## ■ Class Exercise 3

$$\begin{aligned} & \text{minimize} && x^2 + y^2 \\ & \text{subject to} && x^2 + y^2 \leq 5, \quad x + 2y = 4, \quad x, y \geq 0. \end{aligned}$$

# Lagrangian Duality

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# Motivation for Duality

## Primal problem

$$\min_{\mathbf{x}} \max_{\lambda \geq 0, \nu} L(\mathbf{x}, \lambda, \nu)$$

## Dual problem

$$\max_{\lambda \geq 0, \nu} \min_{\mathbf{x}} L(\mathbf{x}, \lambda, \nu)$$

- For any  $(\lambda, \nu)$ :

$$L(\mathbf{x}, \lambda, \nu) \leq \max_{\lambda' \geq 0, \nu'} L(\mathbf{x}, \lambda', \nu').$$

- Therefore:

$$\min_{\mathbf{x}} L(\mathbf{x}, \lambda, \nu) \leq \min_{\mathbf{x}} \max_{\lambda' \geq 0, \nu'} L(\mathbf{x}, \lambda', \nu').$$

- Hence (**weak duality**):

$$\max_{\lambda \geq 0, \nu} \min_{\mathbf{x}} L \leq \min_{\mathbf{x}} \max_{\lambda \geq 0, \nu} L.$$

## Motivation for Duality (continued)

- The solution to the dual problem provides a **lower bound** to the primal optimum.

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^\top \mathbf{g}(\mathbf{x}) + \boldsymbol{\nu}^\top \mathbf{h}(\mathbf{x}).$$

$$\begin{array}{ll} \text{minimize} & f(\mathbf{x}) \\ \text{subject to} & \mathbf{g}(\mathbf{x}) \leq 0, \quad \mathbf{h}(\mathbf{x}) = 0. \end{array}$$



# The Lagrangian Duality

- **Primal problem (P):**

$$\begin{aligned} & \text{minimize } f(\mathbf{x}) \\ & \text{subject to } g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, l, \\ & \quad \quad \quad h_j(\mathbf{x}) = 0, \quad j = 1, \dots, m. \end{aligned}$$

- **Lagrangian Dual Problem (D):**

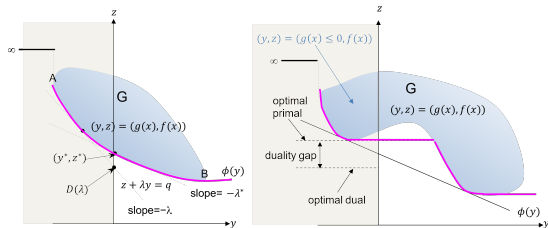
$$\begin{aligned} & \text{maximize } \mathcal{D}(\boldsymbol{\lambda}, \boldsymbol{\nu}) \triangleq \min_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \\ & \text{subject to } \lambda_i \geq 0, \quad i = 1, \dots, l, \end{aligned}$$

where

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \sum_i \lambda_i g_i(\mathbf{x}) + \sum_j \nu_j h_j(\mathbf{x}).$$

## (\*) Geometric Interpretation of Lagrangian Duality

- Consider the primal problem with one inequality constraint:  $\min f(x)$  s.t.  $g(x) \leq 0$ .
- In the  $(y, z)$ -plane, let  $G = \{(y, z) : y = g(x), z = f(x)\}$ . The primal problem finds a point in  $G$  with  $y \leq 0$  that minimizes  $z$ .
- To compute  $\mathcal{D}(\lambda)$  for  $\lambda \geq 0$ : minimize  $z + \lambda y = f(x) + \lambda g(x)$  over points in  $G$ .
- The dual problem is equivalent to finding the slope  $\lambda$  of the supporting hyperplane to  $G$  whose  $z$ -intercept is maximal.



# Weak Duality Theorem

## Weak Duality Theorem

Let  $\mathbf{x}$  be feasible for (P) and  $(\boldsymbol{\lambda}, \boldsymbol{\nu})$  be feasible for (D) (i.e.,  $\boldsymbol{\lambda} \geq \mathbf{0}$ ). Then

$$f(\mathbf{x}) \geq \mathcal{D}(\boldsymbol{\lambda}, \boldsymbol{\nu}).$$

Moreover, if  $f(\mathbf{x}^*) = \mathcal{D}(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ , then  $\mathbf{x}^*$  and  $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$  are optimal for (P) and (D) respectively.

**Proof.** Since  $\mathcal{D}(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_y L(y, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$  and  $\boldsymbol{\lambda} \geq \mathbf{0}$ :

$$\mathcal{D}(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f(\mathbf{x}) + \underbrace{\boldsymbol{\lambda}^\top \mathbf{g}(\mathbf{x})}_{\leq 0} + \underbrace{\boldsymbol{\nu}^\top \mathbf{h}(\mathbf{x})}_{=0} \leq f(\mathbf{x}). \quad \square$$

## Duality Gap

If the optimal objective value of the primal problem is *strictly greater* than the optimal objective value of the dual problem, a **duality gap** is said to exist:

$$f^* - \mathcal{D}^* > 0.$$

# Strong Duality Theorem

## Strong Duality Theorem

Let  $f, g_i$  be convex ( $i = 1, \dots, l$ ) and  $h_j$  be affine ( $j = 1, \dots, m$ ). Suppose **Slater's condition** holds:  $\exists x \in \mathbb{R}^n$  such that  $g_i(x) < 0$  and  $h_j(x) = 0$ . Then

$$\inf\{f(x) : g_i(x) \leq 0, h_j(x) = 0\} = \sup\{\mathcal{D}(\lambda, \nu) : \lambda \geq 0\}.$$

If the inf is finite, the sup is achieved at some  $(\lambda^*, \nu^*)$  with  $\lambda^* \geq 0$ . If the inf is achieved at  $x^*$ , then  $\lambda_i^* g_i(x^*) = 0$  for all  $i$ .

- The Lagrangian  $L(x, \lambda, \nu)$  is convex in  $x$  under these conditions.

## Example: Strong Duality

- Consider:

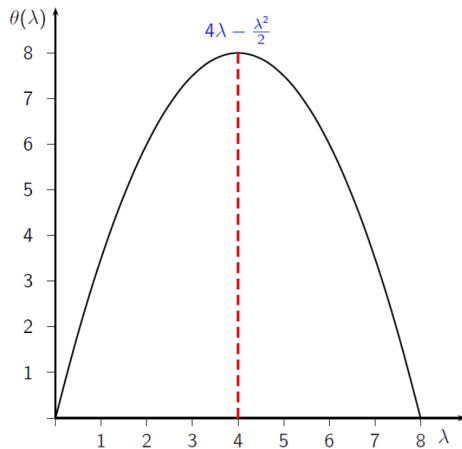
$$\begin{aligned} &\text{minimize } x_1^2 + x_2^2 \\ &\text{subject to } x_1 + x_2 \geq 4, \quad x_1, x_2 \geq 0. \end{aligned}$$

Let  $X = \mathbb{R}_+^2$ . Lagrangian:

$$L(x, \lambda) = x_1^2 + x_2^2 + \lambda(4 - x_1 - x_2).$$

Dual function:

$$\theta(\lambda) = 4\lambda + \min_{x_1 \geq 0} \{x_1^2 - \lambda x_1\} + \min_{x_2 \geq 0} \{x_2^2 - \lambda x_2\}.$$



Graph of the dual function  $\theta(\lambda)$

## Example: Strong Duality (continued)

The minimum of  $L(x, \lambda)$  over  $X = \mathbb{R}_+^2$  is attained at

$$x_1(\lambda) = \frac{\lambda}{2}, \quad x_2(\lambda) = \frac{\lambda}{2}.$$

Thus:

$$\theta(\lambda) = L(x(\lambda), \lambda) = 4\lambda - \frac{\lambda^2}{2} \quad \forall \lambda \geq 0.$$

The dual function is concave and differentiable. Maximize:

$$\frac{\partial \theta}{\partial \lambda} = 4 - \lambda = 0 \Rightarrow \lambda^* = 4.$$

Therefore  $\theta(\lambda^*) = 8$  and  $x^* = x(\lambda^*) = (2, 2)$ . No duality gap:  $f(x^*) = 8 = \theta(\lambda^*)$ . ✓

- **Class Exercise**

- Solve the problem.
- Formulate and solve the dual problem.
- Check whether the primal and dual objective values are equal (strong duality).

$$\begin{array}{ll}\text{minimize} & x_1 \\ \text{subject to} & x_1^2 + x_2^2 = 1.\end{array}$$



- **Primal LP:**

$$\begin{aligned} & \text{minimize } c^T x \\ & \text{subject to } Ax = b, \quad x \geq 0. \end{aligned}$$

- Lagrangian:  $L(x, \lambda, \nu) = c^T x - \lambda^T x + \nu^T (b - Ax)$ .
- Stationarity for the dual:  $c - \lambda - A^T \nu = 0, \lambda \geq 0$ .

- **Dual LP:**

$$\begin{aligned} & \text{maximize } b^T \nu \\ & \text{subject to } A^T \nu \leq c. \end{aligned}$$

# Duality in Convex Problems: Quadratic Programming

- **Primal QP** ( $H \succeq 0$ ):

$$\begin{aligned} & \text{minimize } \frac{1}{2}x^T Hx + d^T x \\ & \text{subject to } Ax \leq b. \end{aligned}$$

- Lagrangian:  $L(x, \lambda) = \frac{1}{2}x^T Hx + d^T x + \lambda^T (Ax - b)$ .
- Stationarity:  $Hx + A^T \lambda + d = 0 \Rightarrow x = -H^{-1}(d + A^T \lambda)$ .
- **Dual QP:**

$$\begin{aligned} & \text{maximize } \frac{1}{2}\lambda^T D\lambda + \lambda^T c - \frac{1}{2}d^T H^{-1}d \\ & \text{subject to } \lambda \geq 0, \end{aligned}$$

where  $D = -AH^{-1}A^T$  and  $c = -b - AH^{-1}d$ .

- This maximizes a *concave* quadratic over the nonnegative orthant.